

Master Power Brakes FAQs

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Disc Brakes FAQ

Why are disc brakes better than drum brakes?

First it must be understood what a brake system does. When a brake pedal is applied, a hydraulic event takes place that causes a friction material to come in contact with a rotor or drum's machined surface. At this point, the vehicle's movement through that drum or rotor is converted into heat. That transfer from motion to heat is what stops a vehicle.

Understanding that process is required to understand why [disc brakes are better than drum brakes](#). A drum brake actually works quite well from the simple standpoint that a brake shoe wedging inside of the drum actually can work better than two brake pads squeezing on a rotor. Unfortunately, a drum brake can not dissipate heat nearly as quick as a disc brake. Therefore, disc brakes are the preferred braking system for most vehicles due simply to efficiency. A brake rotor with its internal vein structure actually pulls air through from the center of the rotor and through the veins to dissipate heat from the brake system.

Which calipers are better – 4 piston or single piston?

There really isn't an answer to this question that will always apply. The key is making sure there is ample clamping force as provided by the caliper and the brake system for the given vehicle. Just because a caliper has more pistons doesn't necessarily make it better or worse.

How much pressure is needed for disc brakes?

This is a very difficult question to answer as the amount of pressure

needed is going to depend on what is being asked of the brake system. If you are slowly braking a vehicle coming up to a red light or stop sign, a much lower but consistent line pressure is needed. This could be somewhere around 750 psi. If the situation is a panic stop from a higher speed, then upwards of 1,500 psi of line pressure might be required.

Can I use my drum brake master for disc brakes?

It is possible. The first thing to check is too make sure the master cylinder can provide the proper volume for the brake system. The other thing to check is too see if the master cylinder has a residual valve. If it does have a residual valve it will require a change to a different master cylinder as the residual valve will hold line pressure and cause the brakes to drag.

Can I run disc brakes with a manual brake master?

Yes. The bore size of the master can not be larger than 1" and your leverage at the pedal has to be greater than power brakes.

If I add disc brakes will I need a proportioning valve?

Yes. The addition of a proportioning valve is a must. Without it your braking system will not operate properly and safely.

Can I re-use the old drum brake valve when I change over to disc brakes?

It is not recommended. A typical four wheel drum brake valve is usually not much more than a distribution block and doesn't provide the proper features that a disc brake system would need.

I have converted my car to disc brakes and now I can't get a pedal. It goes to the floor no matter how much I bleed it.

Check the bleeder screw orientation on the calipers. The bleeder screw must be on top in the 12:00 position. If it is not you will not be able to remove all the air from the system.

[Download Our BRAND NEW Printable Troubleshooting Flow Chart](#)

Master Cylinders FAQ

How does a disc brake master differ from a drum brake master?

A drum brake master will differ from a disc brake master in two ways. The amount of fluid that a drum brake master has to move is less than disc brake. Drum brake masters have 10 lb residual valves at the outlet to keep a residual pressure on the drums. If you use a drum brake master for disc brakes you would move an insufficient volume of fluid and the disc brakes would drag because of the residual valves.

What will happen if I use a master cylinder for drum brakes and it doesn't have a residual valve?

You will have a spongy pedal and you'll have to pump the pedal to get good brakes.

I have manual brakes and I have an extremely hard pedal. Why?

Check the bore size of your master. If it is larger than 1" then you will have a very hard pedal.

What bore size do you need for manual brakes?

Usually a bore size of 1" or smaller is recommended. This is due to the fact that a smaller bore master cylinder will provide more travel in the pedal than a larger bore master cylinder. Therefore, more effort will be required if too large a bore master cylinder is used. The smaller bore master cylinder will provide the same line pressure at a lower pedal effort.

What is the difference between a power brake master and a manual brake master?

A power brake master cylinder will normally have a larger bore diameter than a manual master cylinder. This is of course depending on the caliper used on the vehicle.

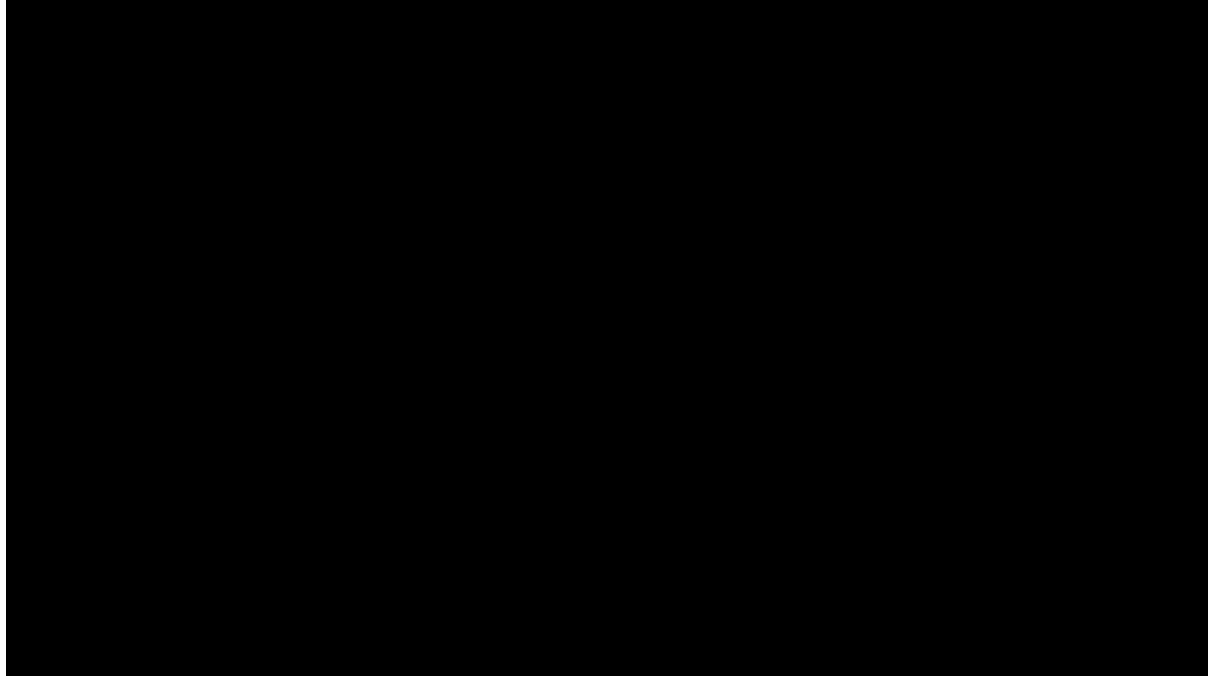
Can I use a shallow hole master in a manual brake application?

It is not recommended. The piston hole must be deep so the manual push rod does not fall out.

Can I use my manual master on a booster?

Yes you can. Keep one thing in mind though. Generally a manual master cylinder will have a deeper hole in the piston. Because of that, a slug in the back of the master cylinder may be required or the use of a longer

check the piston to pin depth between the booster and the master cylinder. A [*Booster Pin Depth Gauge \(p/n: AC2003\)*](#) can be used to check this. For more help, check our [*Blog Post*](#) on this subject or the video below.



Can I use a power brake master without a booster?

Usually not. The bore size will be too large giving a hard pedal and the piston hole will be shallow allowing the push rod to fall out.

Why is one fluid chamber larger than the other in a disc/drum master?

Generally for wear reasons. As disc brake pads wear, the caliper piston(s) will move outward to be closer to the rotor. More fluid is required when this happens and will cause the fluid level in the master cylinder will drop more rapidly than the drum portion so you need more reserve.

Should I buy a re-manufactured or new master?

Always try to use a new master. Re-manufactured master cylinders tend to have a high failure rate. If the only available option is to go rebuilt or have one rebuilt, the best way is to have a stainless steel sleeve installed in the bore.

How can I tell if my master cylinder is bad?

Usually the system has a pedal that is very spongy and feels like you are

pressing on a pile of pillows. Also, if you hold your foot on the brake pedal it will usually sink slowly toward the floor if it is bad.

Why should I eliminate my single piston master?

Safety. By going to a split system the possibility of a complete brake failure is virtually eliminated.

Can I rebuild my master myself?

Yes, but you must have the correct rebuilding kit. Also check the bore to be sure it is not pitted or corroded.

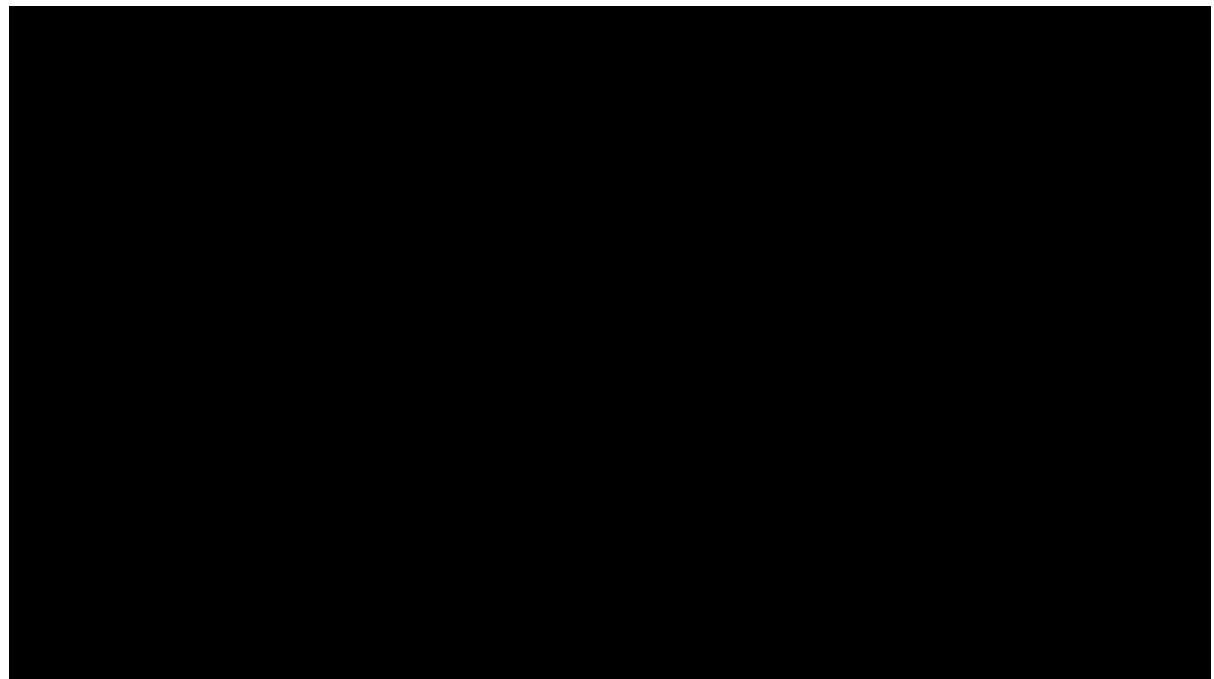
I purchased a replacement master and It does not look like the original.

Can I use it?

More than likely, yes. Many aftermarket replacement masters will have a different casting look but be the same internally.

Do I need to bench bleed the master cylinder before installation on the vehicle?

It is not an absolutely mandatory thing but it is highly recommended. The reason to do it is the master cylinder will usually trap a fair amount of air in it. To bleed it on the car will require a lot of extra effort that isn't necessary. We recommend using our [*Syringe Bleeder Kit \(p/n: AC20010K\)*](#) as it makes the job a much simpler one.



Is a power booster required with disc brakes?

No. Although a power booster will provide a pedal feel that feels better when applied, manual brakes work fine.

What are the symptoms of a bad power booster?

A bad power booster will give a very hard pedal and will feel like you need two feet to stop the car.

What does a booster do?

A power brake booster helps assist pushing the master cylinder piston when you apply the brakes.

How much pressure should I be getting to the wheels with a power booster?

Typically you should expect about 1000 psi. to the wheels for a disc brake system. A disc brake system requires this amount of pressure so be careful when using a smaller 7" booster that puts out only 900 psi pressure. Use our [Brake Pressure Gauge Kit \(p/n:AC2010K\)](#) to get a pressure reading at any point in the brake system.

What size booster do I need?

The key thing to remember with a vacuum brake booster and disc brakes is you can't have too much booster. Four wheel drum brake cars are a different scenario where you can have too much. Don't go with a smaller brake booster just because it fits better or looks cool. Using a booster that is too small will actually create a driving condition that is worse than the worst manual brakes. For a reference point, a musclecar will work just fine with an 8" dual diaphragm booster.

Can I use a Ford Booster on a GM vehicle?

If you can mount the booster properly and link it to the pedal then any booster can be used on any type of car.

How much vacuum is needed to operate a booster properly?

For a power booster to function properly you will need at least 18" of vacuum at idle in park. The further away from 18" the vacuum gets, the harder the brake pedal becomes.

What happens if the vacuum is too low?

If your vacuum level is too low you will experience a hard pedal and it will feel like the vehicle won't stop.

I have a radical cam. Is there any way to supplement my low engine vacuum?

Yes. We offer a [*Silent Drive Vacuum Pump*](#) (p/n: AC9001K) that will supply a constant 18" of vacuum to the brake booster at all times. It is a simple installation requiring a 12-volt connection and can be mounted anywhere within the vehicle.

Will a reserve canister help?

Somewhat, but usually they will only show a moderate increase in braking performance since the engine doesn't really get to a sufficient vacuum level on a constant basis. You really need a constant 18" at least.

What are my options if I don't want to use the vacuum pump?

Not many. Without the supplemental pump the only choice is to eliminate the booster and go to manual brakes or some form of hydroboost.

How will I know if my booster is bad?

A bad booster will give a very hard pedal and it will feel like the vehicle will not stop.

How do I check to see if my booster is operating properly?

Shut off the engine. Depress the brake pedal a few times to evacuate the booster. Apply a steady pressure to the pedal and start the engine. The pedal should fall slightly.

Is there any way to decrease booster size and still maintain assist?

Yes. By using a dual diaphragm design booster you can maintain a high level of assist while keeping a small size. Master Power Brakes offers an 8" dual diaphragm booster that fits most street rod and GM applications.

I have installed a big block and the booster hits the valve covers. Is there anything I can do?

We offer an 8" dual diaphragm booster that will work for most GM and Ford applications.

Will I need a special booster with 4 wheel disc brakes?

The main consideration with 4 wheel disc brakes is that you have plenty of assist as provided by the power brake booster. Don't undersize your booster with a 4 wheel disc set up. The larger and more diaphragms that

When I step on the brakes the pedal feels good but I get no braking.

What could cause this?

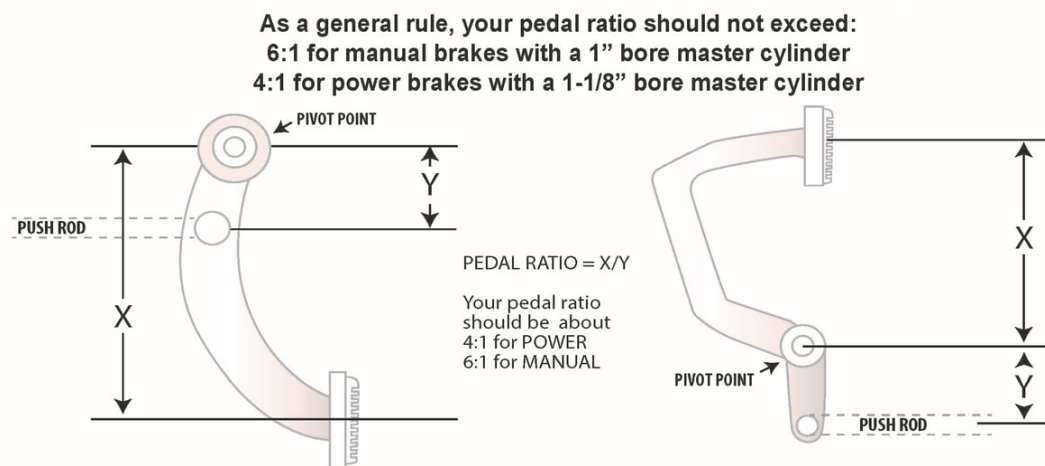
One of the causes of this is a mismatch between the booster pin length and the depth of the master cylinder piston hole. Be sure the hole is not too deep for the booster pin.

I have installed a new booster on my car and now the brakes drag. Why?

Again this can be due to a mismatch between the booster pin and the master cylinder piston. Too long a pin will cause this.

I have installed power brakes on my original manual brake car and now the brakes are extremely sensitive. Why?

The pedal ratio is probably wrong. A power brake vehicle needs a pedal ratio of 4:1. Many manual brake cars were a 6:1 ratio. On some applications, there are two attachment points on the brake pedal. For power brakes you need to use the lower attachment hole. Using the upper hole will make the brakes too sensitive. There are many cars however that will use the same hole for both power and manual as the power brake booster will have a bellcrank as part of the mounting brackets that will change the vehicle to the proper ratio.



Power boosters keep going bad on my vehicles. Why?

You are probably getting corrosive vapors back into the booster. Install a vapor trap and that should eliminate the problem.

Can I rebuild my own booster?

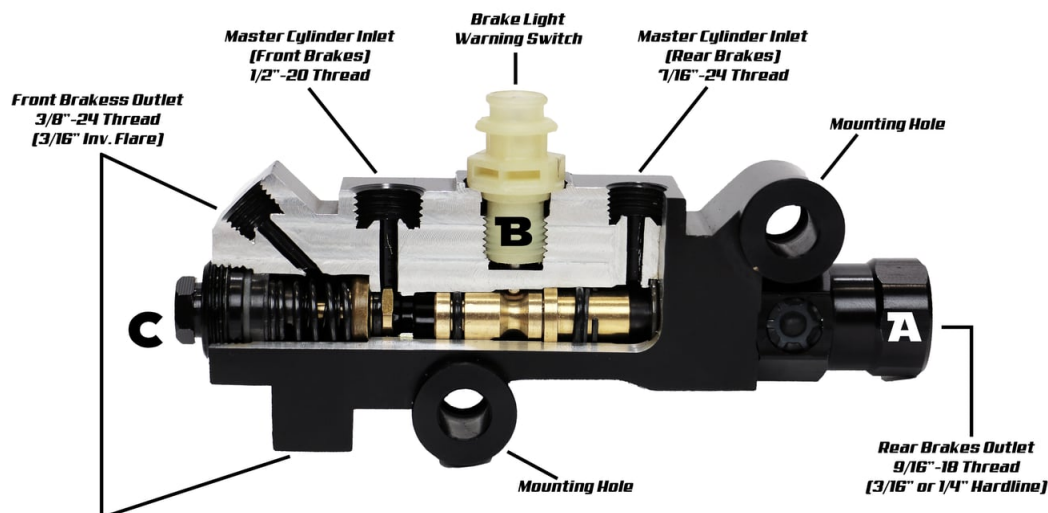
This is not a good idea. There are many parts inside the booster that will require special tools to assemble and re-assemble.

What is a Combination Valve?

A Combination Valve is just what the name implies...a valve that does a combination of things.

Many people will call this particular valve a Proportioning Valve. That isn't necessarily 100% incorrect as a portion of the valve does perform that function. The proper and technical name for this valve is a Combination Valve. We should discuss why it is called a Combination Valve. As we mentioned earlier, this valve does a combination of things and is made up of several different segments doing many different things. There is the Pressure Differential Switch and Valve, the Metering Valve and the Proportioning Valve.

For a look inside of a Combination Valve, check out the picture below:



What does the proportioning part of a Combination Valve do?

The Proportioning Valve part of the Combination Valve (seen above and labeled as A) is used to control the rear brakes. What the Proportioning Valve does and when it actually works is one of the biggest misunderstandings in a Combination Valve. In a moderate braking situation, line pressure being generated by the master cylinder is less extreme and therefore will reach the calipers or wheel cylinders in full. However, when the brakes are really put to work and a panic stop happens, this is when the Proportioning Valve goes to work. In a panic stop, the Proportioning Valve will actually reduce the line pressure to the rear brakes in an effort to try and minimize rear brake lock-up and also the possibility of the rear of the car passing the front of the car which is never a situation anyone wants to experience.

What does the Metering Valve do?

At the front of the Combination Valve (seen above and labeled as C) if using a Combination Valve designated as disc/drum is the Metering Valve. The Metering Valve, sometimes referred to as a Hold-Off Valve, is used to regulate the pressure going to the front wheels when the brakes are initially pressed in a brake system using disc brakes in the front and drum brakes in the rear. Due to the design of disc brakes versus drum brakes, disc brakes will apply quicker than drum brakes. If this happens, the car will nose dive when applying the brakes. To prevent this, the use of a Metering Valve is required. It will simply hold off the line pressure coming from the master cylinder ever so slightly allowing the rear brakes to apply first. The total time it holds off the front brakes is so slight that as a driver, it is never even realized.

The Metering Valve portion becomes less of a discussion when you have a vehicle with disc brakes on the front and rear. In this situation, you do not need a Metering Valve and therefore would by a Combination Valve designated as disc/disc. These valves basically have an open cavity internally in the front that basically allows equal line pressure at equal times to both the front and rear.

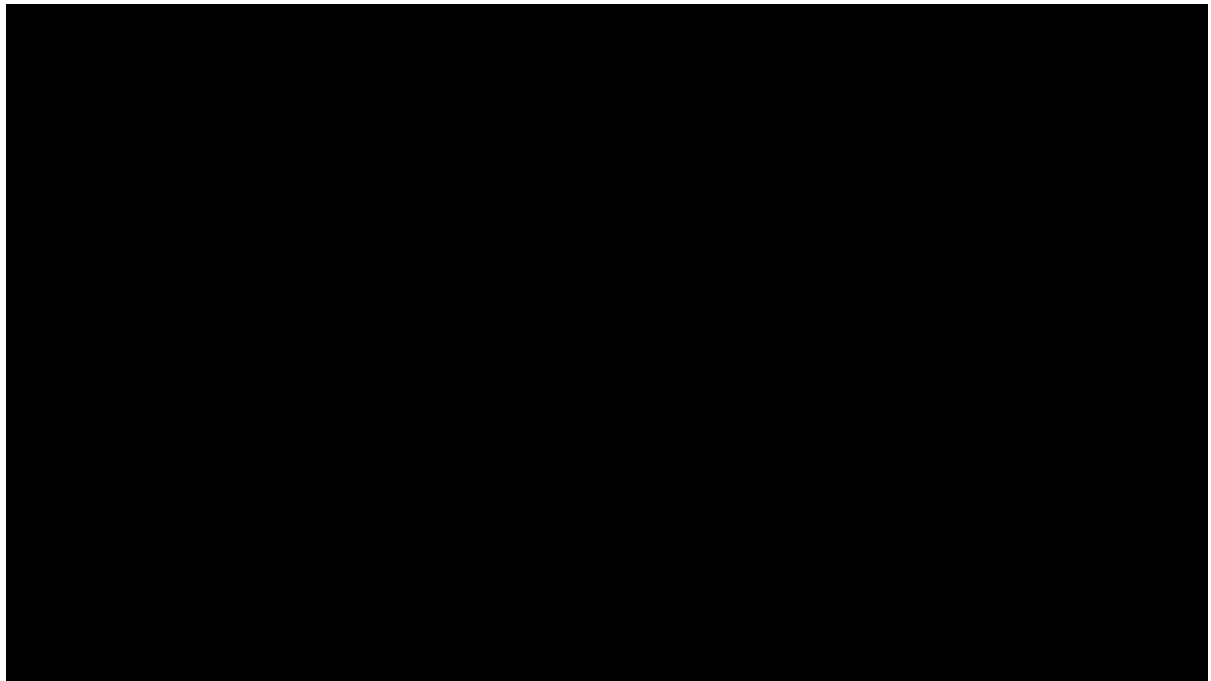
In some instances, a Metering Valve is needed as a stand-alone item. This can be a possibility in a disc/drum vehicle when an Adjustable Proportioning Valve is being used. For this situation, we recommend the use of our [Metering Valve \(p/n: VL3526K\)](#).

What does it mean when a Combination Valve is tripped?

The center portion of the Combination Valve is a safety feature. It is called the Pressure Differential Switch or Valve. It's purpose is to alert a driver if the brake system should lose pressure on either the front or the rear part of the system. Should a pressure loss happen, the balance will be thrown off and allow the valve to move. When it moves, or when some people call it "tripped", the internal valve moves and blocks the non-working port allowing a person to get home.

The bleeding of the brakes can also cause the valve to become tripped. To un-trip a valve, go to the opposite end of the vehicle that might be tripped to and after pumping the brakes and holding them, open a bleeder screw to try and get the center valve to move in the opposite direction. Once the line pressure becomes equal on both sides, the valve

To avoid this from happening when bleeding the brakes, use our [*Bleeder Tool \(p/n: HW3350\)*](#). See the video below to understand this fully.



What does a Residual Valve do?

There are two different uses for residual valves. The 10 lb valve is used to hold a residual pressure to the drum brakes to keep the shoes out close to the drums giving a higher firmer pedal. The 2 lb valve is used in the disc when the master cylinder is lower than the calipers to prevent back siphoning of the fluid from the master. Our residual valves are available as p/n: VL3702K (2lb) or VL3710K (10lb).

What are the symptoms of a bad residual valve?

The brakes will be very spongy and you will need to pump the pedal to get good brakes.

Should I use an adjustable proportioning valve?

We typically would push a customer towards a VL3359K (disc/disc) or VL3360K (disc/drum) Combination Valve but some situations call for an Adjustable Proportioning Valve. The adjustable valve will only provide the proportioning function and not the metering that is needed. The biggest thing to remember is you will use one or the other and never both and if you do elect to use an Adjustable Proportioning Valve, only

install it in the rear brake line on the vehicle. Never install one in the front line of the vehicle.

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